

Cosc 30: Discrete Mathematics

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Problem 1. A distance-2 coloring on an undirected graph G is a labeling $\ell(\cdot)$ on the vertices of G such that every pair of vertices within distance 2 has different labels. In other words, we are looking for a function

$$\ell : V_G \rightarrow \{1, \dots, k\}$$

so that if two vertices x and y share a common neighbor z , then $\ell(x) \neq \ell(y)$; the number of colors used is k .

Prove that every tree whose internal nodes have degree at most 3 has a distance-2 coloring using at most 4 colors.

Solution.

We will prove the statement by induction on the number of nodes in the tree. Let T be an arbitrary tree with n nodes and internal nodes having degree at most 3. Assume for any tree T' with less than n nodes and internal nodes having degree at most 3, T' has a distance-2 coloring using at most 4 colors.

- If T has 1 or 2 nodes we can color it with 4 colors.
- If T has ≥ 3 nodes. Take a leaf node n connected to an internal node v . Note that v is connected to at most 2 other nodes since every internal node has degree at most 3. Remove n to get a tree with less than n nodes, and the internal nodes still have degree at most 3. This has a distance-2 coloring using at most 4 colors. Now attach n back to v . Since v is connected to at most 2 other nodes, v and these two nodes use at most 3 colors, leaving us 1 color to color n .

Hence, T is distance-2 colorable with at most 4 colors.

Problem 2. A directed graph G is semi-connected if, for every pair of vertices u and v , either u is reachable from v or v is reachable from u (or both).

Prove that given any directed acyclic graph G , G is semi-connected if and only if there is a path in G passing through every vertex of G .

Solution.

($\implies$) Assume a directed acyclic graph G is semi-connected. Then there exists an elimination ordering of G such that vertex v_k is a source in $D[v_k, \dots, v_n]$. (At the beginning $v_k = v_1$.) That is edges go from v_i to v_j for $i < j$. Also, since G is semi-connected, for any consecutive pair of vertices in this ordering v_i and v_{i+1} , there exists a path either from v_i to v_{i+1} or from v_{i+1} to v_i . Since G is acyclic, the path must be from v_i to v_{i+1} for all i .

Note that the path between v_i and v_{i+1} connects the two vertices via a single edge because they appear consecutively in the elimination order. If there was some vertex u on the path, then it would appear after v_i but before v_{i+1} in the order, and no such path exists. Hence, there exists a path passing through every vertex in G .

($\impliedby$) Assume a directed acyclic graph G has a path passing through every vertex.

$$P = v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_{n-1} \rightarrow v_n$$

Then for any two vertices v_i and v_j (wlog assume $i < j$) then there exists a path

$$v_i \rightarrow v_{i+1} \rightarrow \dots \rightarrow v_j$$

is a subpath of P , hence v_j is reachable from v_i . Hence, G is semi-connected.